Agenda

Moderation Sohrab Akhoundzadeh
Minutes taker Kantuta Holterman Vargas
Those present Jonas Boselie Tarana Niazi
Absent (excused) None
Location of the meeting
Date 05-06-2026 09:00–10:00

Overview of topics

1 Announcements

Sohrab had to leave at 13:00 for an important family gathering yesterday, meaning the other group members had to continue the measurements. He will compensate for the lost hours by staying the full day today.

2 Approval of agenda

No comments

3 Approve previous minutes

We have been working on the measurements and have exported our data sufficiently so far. During our first measurements, we ran through some problems regarding leakage of our nozzle and also issues with the pressure suddenly not increasing to our desired value after we had done most of our measurements. Luckily, we managed to resolve these issues quickly with the help of our supervisor and Paul Kolpakov.

We have been familiarized with the experimental setup and try to work as efficiently as we could. We rotated around the following tasks: one person to stepwise increase the pressure, one or more persons writing down the measured flow rate and pressure per measurement, one person capturing each frame for each measurement and saving them into a file and one person filling up the measurement sample with the fluid mixture (purely distilled water on day 1) and setting the timer for 1 minute. After doing some measurements, we recalibrated our frame such that the breakup point of the jet is moved high enough in our frame so we can do measurements again before recalibrating again. For the first mixture, this process had to be done regularly and we found that this took a big chunk of our time to finish the measurements.

At the end of the lab session, we managed to find the pressure where we achieved the turbulent regime. Since we are only interested in consistent jets for our data, we decided to stop here and move on to the next mixture.

During day 2, we first got our new mixure by mixing 800mL water with 200 mL ethanol. This time our measurements went a lot more efficient and we noted that we didnt have to recalibrate our frame as often as during the first day.

4 Previous Tasks

- Everyone: last week,
  read the literature provided for the project ✓,
  rotated between measuring, calibration of the camera frame, taking notes and noting down the results for the flow rate per measurement ✓,

- Tarana: 03/05/2026,
  exported three file bundles from data ✓, Sohrab: 03/05/2026,
  added Clara Nellist to github repository ✓,

Task (next meeting): Mention goals for project + issues during measurements (pressure, sudden fluctuation in flow rate in measurement 10 at 1250 mbar) (Sohrab)

5 Topic 1

We have learned that there are 3 main forces that compete with each other during the jet: surface tension (1), inertia (2), viscosity (3). Because every jet is imperfect (imperfect shape of nozzle on microscopic scale, fluctuations during stabilized pressure, etc), the surface tension will trigger these disturbances. As a consequence, the energy of the system can't stay concentrated in a perfectly cylindrical jet with an approximated area of. Instead the disturbances will grow over time and cause the cylindrical jet to now break up in spherical droplets having surface area $4\pi r^2$. This is called the Rayleigh-Plateau instability.

There are three different dimensionless numbers that are of particular importance for the breakup length of a jet depending on the specific mixture used. These are: Reynolds numbers (Re), Weber number (Wb) and Ohnesorge number (Oh). Reynolds number measures the inertia over the viscosity, Webers number measures the inertia, which as opposes to Reynolds number now scales more intensely due to the increased contribution of the jet velocity. Ohm's number measures the rate of viscosity over the surface tension and the inertia. The velocity of the fluid is measured by the following relation: $m = \rho u A$, where ρ denotes the fluid's density (g/m^3), u denotes the fluid's velocity (m/min), A the surface area (m^2) and Q the flow rate (g/min). Eventually, the relation between the breakup length of the jet and Webers number for every mixture of fluids is expected to have a decreasing upward relationship.

Task (before the next meeting): The "who, what, when and why questions" still need to be answered in onennote and download current data files to export to github (Sohrab).
Task (before the next meeting): Read off average value from breakup length off current data (Tarana).
Task (before the next meeting): Work on the schedule (Kantuta)
Task (before next week's lecture): Follow instructions on installing Manim and install on desktop/computer (Everyone)

6 Topic 2

There are two kinds of flow rates: volumetic (1): Q = uA and mass flow rate (2): m = ρuA. Change this topic 1.

7 Remaining discussion points

The supervisor told us to correct some things that were mentioned during topic one. Of particular importance were the SI units that have to be corrected as well.

8 Final question round

Questions about research question regarding influence of parameter change on breakup length.

9 Next Meeting

The next meeting will be on Monday 8th of June between 13:00-14:00 in C2.211. Sohrab will be mostly absent on Tuesday 9th of June because of his retake exam.

10 List of Tasks

Sohrab ↪ next meeting
Mention goals for project + issues during measurements (pressure, sudden fluctuation in flow rate in measurement 10 at 1250 mbar) (05-06-2026) . . 3

Sohrab ↪ before the next meeting
The ”who, what, when and why questions” still need to be answered in onennote and download current data files to export to github (05-06-2026) 3

Tarana ↪ before the next meeting
Read off average value from breakup length off current data (05-06-2026) . 3

Kantuta ↪ before the next meeting
Work on the schedule (05-06-2026) 3

Everyone ↪ before next week’s lecture
Follow instructions on installing Manim and install on desktop/computer (05-06-2026) 3

11 Closing the meeting

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